

Jackson Bopp

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Computer Engineering student and systems integrator seeking an entry-level software engineering role, with hands-on experience across full-stack development, AI/LLM integration, and embedded systems.

Skills

Systems Integration: End-to-end systems from microcontroller firmware to full-stack apps and LLM services; cross-layer debugging

AI / LLM Tools: Anthropic Claude API, Google Gemini API, IBM watsonx.ai (Granite), ElevenLabs, Ollama, OpenWakeWord

Languages: Python, TypeScript, JavaScript, C, C++, C#, MATLAB, R, Verilog/VHDL, RISC-V Assembly

Frameworks & Libraries: Next.js, React, FastAPI, PyQt6, Tailwind CSS, NumPy, SciPy, Matplotlib

Tools & Infrastructure: Docker, MQTT, PostgreSQL, MongoDB, SQLite, Git, Cursor, Raspberry Pi, Arduino, Linux

Certifications: Certified SOLIDWORKS Associate (CSWA), CSWA Additive Manufacturing

Education

University of South Florida

Tampa, FL

B.S. Computer Engineering, Dean's List, Gold's Scholars | GPA: 3.43/4.00

Graduating Fall 2026

Projects

Testbench

July 2026

Next.js, TypeScript, Python, PostgreSQL, MQTT, Docker, IBM watsonx.ai

- Built a full-stack hardware test automation platform streaming live telemetry from an MSP430 microcontroller through a Raspberry Pi bridge over MQTT to a Next.js dashboard with real-time SSE updates.
- Integrated IBM watsonx.ai (Granite 3.8B) for AI-powered failure analysis and an embedded, context-aware chat assistant, both backed by a Drizzle ORM data layer and REST API.
- Containerized the stack with Docker Compose (PostgreSQL and Mosquitto), and wrote the MSP430 firmware and Python UART-to-MQTT bridge for the Raspberry Pi.

Edward, Desktop AI Assistant

May 2026

Python, PyQt6, ElevenLabs, faster-whisper, FastAPI, PyAutoGUI, SQLite

- Created a Windows desktop assistant at the IBM Bob Hackathon with a PyQt6 overlay UI, system tray integration, a smart clipboard and context capture system, and an encrypted password vault.
- Wired up ElevenLabs and faster-whisper for a full voice pipeline, plus a FastAPI and PyAutoGUI computer control system for mouse and keyboard automation, including a confirmation handler.

Radiology Second-Opinion Agent

July 2026

FastAPI, Docker Compose, MLflow, Chroma, Evidently, pytest

- Data and MLOps engineer on a 4-person hackathon team building an AI chest X-ray analysis system; owned the FastAPI serving layer, an MLflow registry, a Chroma vector store, Docker Compose orchestration, drift monitoring, and the test suite.

VoiceLegacy

April 2026

Next.js 16, Tailwind CSS, Clerk, MongoDB Atlas, ElevenLabs, Gemini

- Led all frontend development at the Hackabull hackathon: landing page, phrase bank layout with category filters, the Speak For Me AI rewrite flow, onboarding, voice recording page, and mobile layout polish across every page.
- Connected the app to ElevenLabs for voice cloning and text-to-speech, Gemini for a local fallback path, and Clerk for authentication, deployed to Vercel with a MongoDB Atlas backend.

Open Source

Open Source Contributions

August 2026

C++, Python, C, D-Bus, Linux, Gerrit and GitHub code review

- **OpenBMC phosphor-user-manager:** diagnosed and fixed two bugs in a BMC user management daemon, an unhandled exception that could abort account setup for every user over one bad system record, and a C++ virtual dispatch issue causing unit tests to silently run against the real system instead of test mocks; submitted for review with full before and after test verification.
- **Zephyr RTOS:** found and fixed a devicetree macro bug in the MIPI DBI display driver that produced an incorrect chip select GPIO check for out of range devices; submitted with a compile time regression test.
- **PlatformIO:** root-caused and fixed a crash in the serial device monitor triggered by non-interactive input; merged upstream with an added regression test.

Experience

City of Winter Haven Technology Services

April 2026 – Present

Smart City Student Volunteer

Winter Haven, FL

- Installed server hardware and performed rack maintenance for city technology infrastructure.
- Provisioned and configured YubiKey security keys for staff authentication.
- Resolved IT support tickets and optimized PyTorch/TensorFlow environments for the city's early-stage AI hosting infrastructure.

Subs N' Such

April 2023 – March 2026

Manager, Shift Leader, and Front-End Web Support

Lutz, FL

- Developed the front-end of the restaurant's Square-based ordering site, working within the platform's UI constraints to deliver a functional ordering experience.
- Managed shift operations and enforced food safety procedures.

Involvement

Member, Society of Hispanic Professional Engineers (SHPE), Society of Asian Scientists and Engineers (SASE), and IEEE